# TUGAS 02 F1101 MODUL 02

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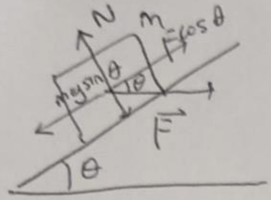
Jawaban ($g = 9{,}8\ m/s^2$)

4.



$m = 100$ kg
$\theta = 30^\circ$
$\mu_k, \mu_0 = 0$

(a) $\Sigma F_x = 0$

$\Rightarrow F\cos\theta - mg\sin\theta = 0 \Rightarrow F = \frac{mg\sin\theta}{\cos\theta}$

$\Rightarrow F = mg \cdot \tan\theta$

$= 100 \cdot 9{,}8 \cdot \tan 30^\circ$

$= 980 \cdot \frac{1}{3}\sqrt{3}$

$\approx \boxed{565{,}80326\ N}$

(b) $\Sigma F_y = 0$

$\Rightarrow N - mg\cos\theta - F\sin\theta = 0$

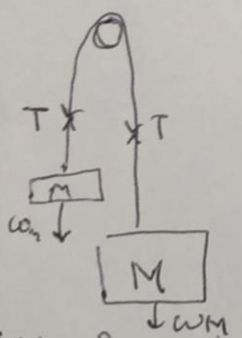
$\Rightarrow N = mg\cos\theta + F\sin\theta$

$= 980 \cdot \cos(30^\circ) + 980 \cdot \frac{1}{3}\sqrt{3} \cdot \sin(30^\circ)$

$= 980 \cdot \frac{1}{2}\sqrt{3} + \frac{980}{2} \cdot \frac{1}{3}\sqrt{3}$

$\approx \boxed{1131{,}60652\ N}$

5.



$m_m = 10$ kg, m = monyet
$m_M = 15$ kg M = peti
"katrol" tak bermassa

(a) Agar bisa mengangkat, gaya yang diberikan monyet = peti

$\Rightarrow F_{monyet} = F_{peti}$

$\Rightarrow m_m \cdot a_{min} + m_m \cdot g = m_M \cdot g$

$\Rightarrow a_{min} = \frac{m_M \cdot g - m_m \cdot g}{m_m} = \frac{15 \cdot 9{,}8 - 10 \cdot 9{,}8}{10} = \boxed{4{,}9\ m/s^2}$

(b) $T - w_m = m_m \cdot a_m$ (1)

$\Rightarrow w_M - T = m_M \cdot a_M$ (2)

Karena satu sistem, $a_m = a_M$

$\Rightarrow \dfrac{T - w_m}{m_m} = \dfrac{w_M - T}{m_M}$

$\Rightarrow \dfrac{T - 98}{10} = \dfrac{15 \cdot 9{,}8 - T}{15} \Rightarrow T = 117{,}6\ N$

Dari persamaan (1), $a_m = \dfrac{T - w_m}{m_m}$

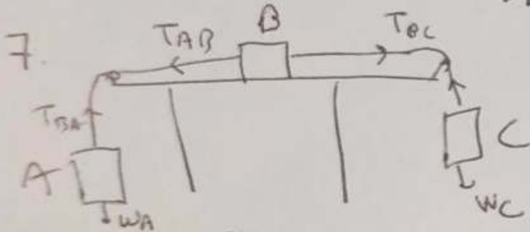
$= \dfrac{117{,}6 - 98}{10}$

$= \boxed{1{,}96\ m/s^2}$

(c) Karena $m_m < m_M$, monyet terangkat ke atas, maka $\boxed{\text{percepatan ke atas}}$ ↑

(d) Menurut (b), $\boxed{T = 117{,}6\ N}$

7.




Licin, $m_A = 6\ kg$

$m_B = 8\ kg$

$m_C = 10\ kg$

$T_{BC}$?

Karena $w_C > w_A$, sistem ke kanan

$w_C - T_{BC} = m_C \cdot a$ (i)

$\dfrac{w_C - w_A}{m_A + m_B + m_C} = a$ (ii)

$\Rightarrow a = \dfrac{98 - 58{,}8}{24} = 1{,}63\overline{3}\ m/s^2$

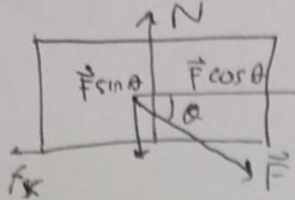
Substitusi ke persamaan (i), didapat

$98 - T_{BC} = 10 \cdot 1{,}6\overline{33}$

$\Rightarrow T_{BC} = 98 - 16{,}\overline{33}$

$= \boxed{81{,}\overline{66}\ N}$

9.




$m = 3{,}5\ kg$ $\mu_K = 0{,}25$

$|\vec{F}| = 15\ N$

$\theta = 40^\circ$

(a) $f_K = \mu_K . N$

$= 0{,}25 . (mg + F \sin\theta)$

$= 0{,}25 . (3{,}5 . 9{,}8 + 15 . \sin 40^\circ)$

$\approx 0{,}25 (43{,}9418)$

$\approx \boxed{10.9854\ N}$

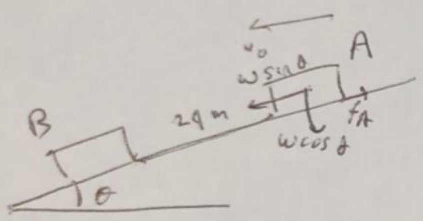
(b) $ma = F\cos\theta - f_K$

$\Rightarrow a = \dfrac{F\cos\theta - f_K}{m}$

$\approx \dfrac{15 . \cos 40^\circ - 10{,}9854}{3{,}5}$

$\approx \boxed{0{,}1442\ m/s^2}$

10.




$\theta = 12^\circ$

$d = 24\ m$

$V_0 = 18{,}0\ m/s$

(a) $\mu_K = 0{,}6$

$f_A = \mu_K . N$

$= 0{,}6 (m_A g . \cos\theta)$

$= 0{,}6 . m_A\ 9{,}8 . \cos 12^\circ$

$\approx 5{,}7515\ m_A$

$w\sin\theta - f_A = m_A . a$

$\Rightarrow a = \dfrac{w\sin\theta - f_A}{m_A}$

$= \dfrac{9{,}8 m_A . \sin 12^\circ - 5{,}7515 m_A}{m_A}$

$= -3{,}71\ m/s^2$

$v_f^2 = v_0^2 + 2ad$

$\Rightarrow v_f = \sqrt{324 - 178{,}08}$

$= \sqrt{145{,}92}$

$\approx \boxed{12{,}08\ m/s}$

(b) $\mu_K = 0{,}1$

Pakai langsung mencari percepatan

$a = g(\sin\theta - \mu_K . \cos\theta)$

$= 9{,}8 . (\sin 12^\circ - 0{,}1 . \cos 12^\circ)$

$\approx 9{,}8 . 0{,}1101$

$\approx 1{,}08\ m/s^2$

$v_f^2 = v_0^2 + 2ad$

$\Rightarrow v_f = \sqrt{324 + 51{,}84}$

$= \sqrt{375{,}84}$

$\approx \boxed{19{,}38\ m/s}$